

because NTV does not exist yet. In 1999, 918 weights (47.4%) are strictly positive, while 1,020 (52.6%) are strictly negative. The negative weights sum to -2.26. $\hat{\sigma}_{fe} = 0.91$: β_{fe} and δ^{TR} may be of opposite signs if the standard deviation of the effect of NTV across subregions is above 0.91 percentage point. $\hat{\sigma}_{fe} = 1.23$: β_{fe} may be of a different sign than the treatment effect in every subregion if the standard deviation of the effect of NTV across subregions is above 1.23 percentage point, a plausible amount of treatment effect heterogeneity.

Therefore, β_{fe} can only receive a causal interpretation if the effect of NTV is constant across subregions, or if the weights attached to it are uncorrelated with the intensity of that effect in each subregion (Assumption 7). These assumptions are not warranted. First, we estimate $\hat{\beta}_{fe}$ again, weighting the regression by subregions' populations. We obtain $\hat{\beta}_{fe} = 14.89$, more than twice its value in the unweighted regression, and the difference between the coefficients is significant (t-stat=2.46). Therefore, we can reject the null that the treatment effect is constant: if the treatment effect was constant across subregions, the weighting would not matter so both the unweighted and the weighted regressions would estimate the same parameter. Second, the weights attached to $\hat{\beta}_{fe}$ are correlated with variables that are likely to be themselves associated with the intensity of the effect in each subregion. For instance, the correlation between the weights and subregions' populations is equal to 0.35 (t-stat=14.01). The effect of NTV may be higher in less populated subregions, as those regions are more rural and fewer other sources of information may be available there. This would lead to a violation of Assumption 7.

3 Extensions of the decomposition results

We consider hereafter several extensions of our decompositions of β_{fe} and β_{fd} in the paper. First, we consider decompositions under the common trends assumption, and under the assumption that the treatment effect is stable over time. Second, we extend our decompositions to ordered treatments. Third, we investigate the effect of including covariates in the regression. Fourth, we study two-way fixed effects 2SLS regressions. Throughout, we focus on sharp designs to ease the exposition. Nevertheless, all the results generalize to fuzzy designs.

3.1 β_{fe} and β_{fd} as weighted sums of ATEs of switching groups

We first show that under an additional condition, β_{fe} and β_{fd} can be written as a weighted sum of the ATEs of switching groups.

Assumption S2 (*Stable treatment effect*) For all g and $t \geq 2$,

$$E(\Delta_{g,t} | \mathbf{D}) D_{g,t-1} = E(\Delta_{g,t-1} | \mathbf{D}) D_{g,t-1}.$$

The stable treatment effect assumption requires that the ATE of every group treated in $t - 1$ does not change from $t - 1$ to t . By iteration, the ATE of a group treated for instance from period t_0 to T is unrestricted before t_0 but should be constant from t_0 to T . Assumption S2 rules out the possibility that the treatment effect changes over time. Therefore, it may be implausible and should be carefully discussed.

We now show that under the common trend and stable treatment effects assumptions, β_{fe} and β_{fd} may identify weighted averages of ATEs. Let $N_S = \sum_{(g,t): D_{g,t} \neq D_{g,t-1}} N_{g,t}$ and, for all g and $t \geq 2$,

$$w_{g,t}^S = \frac{(D_{g,t} - D_{g,t-1}) \sum_{t' \geq t} \frac{N_{g,t'}}{N_{g,t}} \varepsilon_{g,t'}}{\sum_{(g,t): t \geq 2} \left[\frac{N_{g,t}}{N_S} (D_{g,t} - D_{g,t-1}) \sum_{t' \geq t} \frac{N_{g,t'}}{N_{g,t}} \varepsilon_{g,t'} \right]},$$

$$w_{fd,g,t}^S = \frac{(D_{g,t} - D_{g,t-1}) \varepsilon_{fd,g,t}}{\sum_{(g,t): t \geq 2} \frac{N_{g,t}}{N_S} (D_{g,t} - D_{g,t-1}) \varepsilon_{fd,g,t}}.$$

Theorem S2 *Suppose that Assumptions 1-5 and S2 hold. Then,*

$$\beta_{fe} = E \left(\sum_{(g,t): D_{g,t} \neq D_{g,t-1}, t \geq 2} \frac{N_{g,t}}{N_S} w_{g,t}^S \Delta_{g,t} \right),$$

$$\beta_{fd} = E \left(\sum_{(g,t): D_{g,t} \neq D_{g,t-1}, t \geq 2} \frac{N_{g,t}}{N_S} w_{fd,g,t}^S \Delta_{g,t} \right).$$

Moreover, $w_{fd,g,t}^S \geq 0$ for all g and $t \geq 2$. If Assumption 6 holds and $N_{g,t}/N_{g,t-1}$ does not vary across g for all $t \geq 2$, $w_{g,t}^S \geq 0$ for all g and $t \geq 2$.

Theorem S2 shows that in sharp designs, under the common trends and stable treatment effect assumptions, β_{fe} and β_{fd} identify weighted sums of ATEs of switching cells. The weights differ from those in Theorems 1 and 2. Now the weights attached to β_{fe} are all positive in staggered adoption designs, while the weights attached to β_{fd} are all positive in all sharp designs. Therefore, in staggered adoption (resp. sharp) designs, β_{fe} (resp. β_{fd}) relies on the assumption that the treatment effect is stable over time, but it does not require that treatment effects be homogeneous between groups.

3.2 Non-binary, ordered treatment

We now consider the case where the treatment takes a finite number of ordered values, $D_{i,g,t} \in \{0, 1, \dots, \bar{d}\}$, and show that Theorem 1 can easily be extended to this case.⁵ We need to define

⁵Theorem 2 can also be extended to the case of a non-binary treatment.

potential outcomes for all the possible treatment values. For instance $Y_{i,g,t}(\bar{d})$ is the counterfactual outcome of observation i in cell (g, t) if she receives treatment value $\bar{d}$. We also need to modify the treatment effect parameters we consider. In lieu of δ^{TR} , we consider the average causal response (ACR) on the treated,

$$\delta^{ACR} = E \left(\frac{1}{N_1} \sum_{i,g,t} Y_{i,g,t}(D_{g,t}) - Y_{i,g,t}(0) \right).$$

Similarly, for all (g, t) such that $D_{g,t} \neq 0$, we consider, instead of $\Delta_{g,t}$,

$$\Delta_{g,t}^{ACR} = \frac{1}{N_{g,t} D_{g,t}} \sum_{i=1}^{N_{g,t}} [Y_{i,g,t}(D_{g,t}) - Y_{i,g,t}(0)].$$

Then, similarly to (2), the following decomposition holds:

$$\delta^{ACR} = E \left(\sum_{(g,t): D_{g,t} \neq 0} \frac{N_{g,t} D_{g,t}}{N_1} \Delta_{g,t}^{ACR} \right).$$

Let $w_{g,t}^O = \frac{\varepsilon_{g,t}}{\sum_{g,t} \frac{N_{g,t} D_{g,t}}{N_1} \varepsilon_{g,t}}$. Note that if the treatment is binary, $w_{g,t}^O = w_{g,t}$.

Theorem S3 *Suppose that Assumptions 1-5 hold and $D_{i,g,t} \in \{0, \dots, \bar{d}\}$. Then,*

$$\beta_{fe} = E \left(\sum_{(g,t): D_{g,t} \neq 0} \frac{N_{g,t} D_{g,t}}{N_1} w_{g,t}^O \Delta_{g,t}^{ACR} \right).$$

Theorem S3 shows that under Assumption 5, when the treatment is not binary β_{fe} identifies a weighted sum of the ACRs in all the (g, t) cells that are not untreated. Then, since the proof of Corollary 1 does not rely on the nature of the treatment, Corollary 1 directly applies to ordered treatments as well, by just replacing $w_{g,t}$ and $N_{g,t}$ by $w_{g,t}^O$ and $N_{g,t} D_{g,t}$, respectively. Corollary 2 extends as well to this set-up, by simply modifying the no-correlation condition appropriately.

3.3 Including covariates

Often times, researchers also include a vector of covariates $X_{g,t}$ as control variables in their regression. In this section, we show that our Theorem 1 can be extended to this case.⁶ We start by redefining Regression 1 in this context.

⁶Theorem 2 can also be extended to regressions with covariates.

Regression 1X (*Fixed-effects regression with covariates*)

Let $\widehat{\beta}_{fe}^X$ denote the coefficient of $D_{g,t}$ in an OLS regression of $Y_{i,g,t}$ on group and period fixed effects, $D_{g,t}$, and $X_{g,t}$. Let $\beta_{fe}^X = E(\widehat{\beta}_{fe}^X)$.

Then, we need to modify Assumptions 3-5. Hereafter, we let $\mathbf{X}_g = (X_{g,1}, \dots, X_{g,t})$.

Assumption S3 (*Independent groups with covariates*) The vectors $(Y_{g,t}(0), Y_{g,t}(1), D_{g,t}, X_{g,t})_{1 \leq t \leq T}$ are mutually independent.

Assumption S4 (*Strong exogeneity and common trends with covariates*) There is a vector γ of same dimension as $X_{g,t}$ such that

$$E(Y_{g,t}(0) - Y_{g,t-1}(0) - (X_{g,t} - X_{g,t-1})'\gamma | \mathbf{D}_g, \mathbf{X}_g) = E(Y_{g,t}(0) - Y_{g,t-1}(0) - (X_{g,t} - X_{g,t-1})'\gamma)$$

and $E(Y_{g,t}(0) - Y_{g,t-1}(0) - (X_{g,t} - X_{g,t-1})'\gamma)$ does not vary across g .

Rearranging, Assumption S4 requires that

$$E(Y_{g,t}(0) | \mathbf{D}_g, \mathbf{X}_g) - E(Y_{g,t-1}(0) | \mathbf{D}_g, \mathbf{X}_g) = (X_{g,t} - X_{g,t-1})'\gamma + \lambda_t,$$

for some constant λ_t . Then, Assumption S4 allows for the possibility that groups experience different evolutions of their $Y_{g,t}(0)$ over time, but it requires that those differential evolutions are fully accounted for by a linear model in $X_{g,t} - X_{g,t-1}$, the change in a group's covariates. Assumption S4 is implied by the linear model that is often invoked to justify the use of the FE regression with covariates. For instance, the use of Regression 1X is often justified by the following model:

$$Y_{g,t}(0) = \gamma_g + \lambda_t + X'_{g,t}\gamma + \eta_{g,t}, \quad E(\eta_{g,t} | \mathbf{D}_g, \mathbf{X}_g) = 0. \quad (24)$$

Equation (24) implies Assumption S4, but it does not imply Assumption 5.

An interesting special case is when the control variables are group-specific linear trends. Then, Assumption s4 requires that for all $t \geq 2$,

$$E(Y_{g,t}(0) | \mathbf{D}_g, \mathbf{X}_g) - E(Y_{g,t-1}(0) | \mathbf{D}_g, \mathbf{X}_g) = \gamma_g + \lambda_t,$$

for some constants γ_g and λ_t . From $t-1$ to t , the evolution of $Y(0)$ in group g should deviate from its group-specific linear trend γ_g by an amount λ_t common to all groups. Then, Assumption S4 is a “common deviation from linear trends” assumption, which may be more plausible than the standard common trends assumption.

Let $\varepsilon_{g,t}^X$ denote the residual of observations in cell (g, t) in the regression of $D_{g,t}$ on group and period fixed effects and $X_{g,t}$. One can show that if the regressors in Regression 1 are not collinear, the average value of $\varepsilon_{g,t}^X$ across all treated (g, t) cells differs from 0: $\sum_{(g,t): D_{g,t}=1} (N_{g,t}/N_1) \varepsilon_{g,t}^X \neq 0$. Then, let

$$w_{g,t}^X = \frac{\varepsilon_{g,t}^X}{\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} \varepsilon_{g,t}^X}.$$